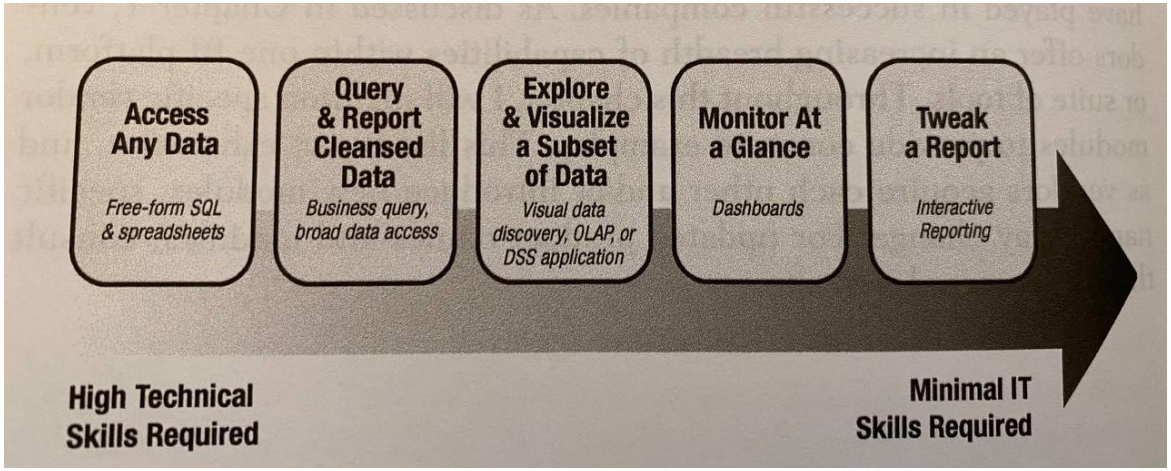
CHAPTER 3:

BUSINESS INTELLIGENCE FRONT-END

Self-Service Business Intelligence

- Offering users self-service BI is top priority for many BI teams.
- The vision is that if users have easy-to-use BI tools,
  - ✓ they can service themselves, asking tough business questions that yield the most value, without any assistance from IT.
  - ✓ could be no IT backlog of request.
  - ✓ users could act at the fast pace business demands.
- In reality, it is a bit different and there are number of variables, including staffing, complexity and tech savviness.

Self-Service Business Intelligence



Self-service BI continuum

- When you think about the vision for self-service BI, consider the use case, sophistication of the users, degree of IT involvement and breadth of data sources.

Business Query and Reporting

- Business query and reporting tools:
  - ✓ Allow business users to create queries and report themselves without rely on IT person and most often used for decision making and management purpose.
  - ✓ Is a key module to provide users with self-service information access.
- When BI first emerged as a category in the early 1990s, it was based on the advent of business query tools that allowed users to create queries without knowing SQL. For the first time, data locked up in relational databases was now accessible to those other than SQL programmers.
- In some cases, a report is truly **ad hoc**; it's a one-off business question that will never be posed again.
- Ad-hoc queries may be exploratory in nature as users try to find the root cause of a problem, test a theory or consider changing a business model.

Business Query and Reporting

Functional Area	Fixed Report	Purpose	Related Ad Hoc Query
Supply Chain	Inventory by Product	To determine if an order can be fulfilled today by the primary warehouse	If I'm short at my main warehouse, can I supply the product from elsewhere?
Marketing	Top 10 Customers by Quarter and Product	To understand which customers generate the most revenue	Which customers fell off this quarter's list? Are there certain products we can cross-sell?
Supply Chain	Raw Material Receipts and Delivery Times	To determine how long it takes to acquire raw materials and which supplier can fulfill purchase orders fastest	Are there other suppliers who can respond faster?
Management	Patients per Hour	To understand busy periods and wait times	Do staffing levels correspond to busy times?
Accounting	Accounts Receivable Aging	To identify the number of days particular invoices have been open; which customers are greater than 60 days?	Have these customers paid late before? Has a reminder notice been sent?
Human Resources	Average Salary by Job Level	Monitor compensation	Are there differences in pay by tenure, age, or gender?

Sample fixed and ad-hoc reports

Business Query and Reporting

- Fix reports may lead to an ad-hoc query.
- As users explore the data, what started as an ad-hoc query or one-time question may later become a fixed report.
- It's important to recognize the iterative nature of business intelligence and ensure you have flexible business intelligence tools.

Business Query and Reporting

- Getting to the data is just one capability of business query tools; the other aspect is presenting and formatting the data in a meaningful way, loosely referred to as **reporting**.

- The terms “query” and “reporting” are sometimes used interchangeably because a business query and reporting tool will have both capabilities – getting to the data and formatting it to create a report.

Business Query and Reporting

- Business query tool allow business users to access a data source via business terms without having to write any SQL.

- The data source could be a data warehouse or it might be direct access to an operational system.

- A key feature of a business query tool is that it has a business view or metadata layer that hides the complexity of the physical database structure from the business user by:
  - ✓ using business terminology rather than physical field names.
  - ✓ Automatically connecting related tables via joins.
  - ✓ Providing metrics that may calculate and aggregate facts such as revenue, no. of customers, no. of orders, average sales, etc.

Business Query and Reporting

- For integrated BI platforms, the business view is common to all the BI tool modules:
  - ✓ Business query
  - ✓ Reporting‘
  - ✓ Analysis
  - ✓ Dashboard
- When the business view looks too much like the data warehouse or source system with confusing table and field names, business users can too easily build incorrect queries.
- Poor business view design also forces users to put too much logic and too many calculations inside individual reports and dashboards.

Visual Data Discovery

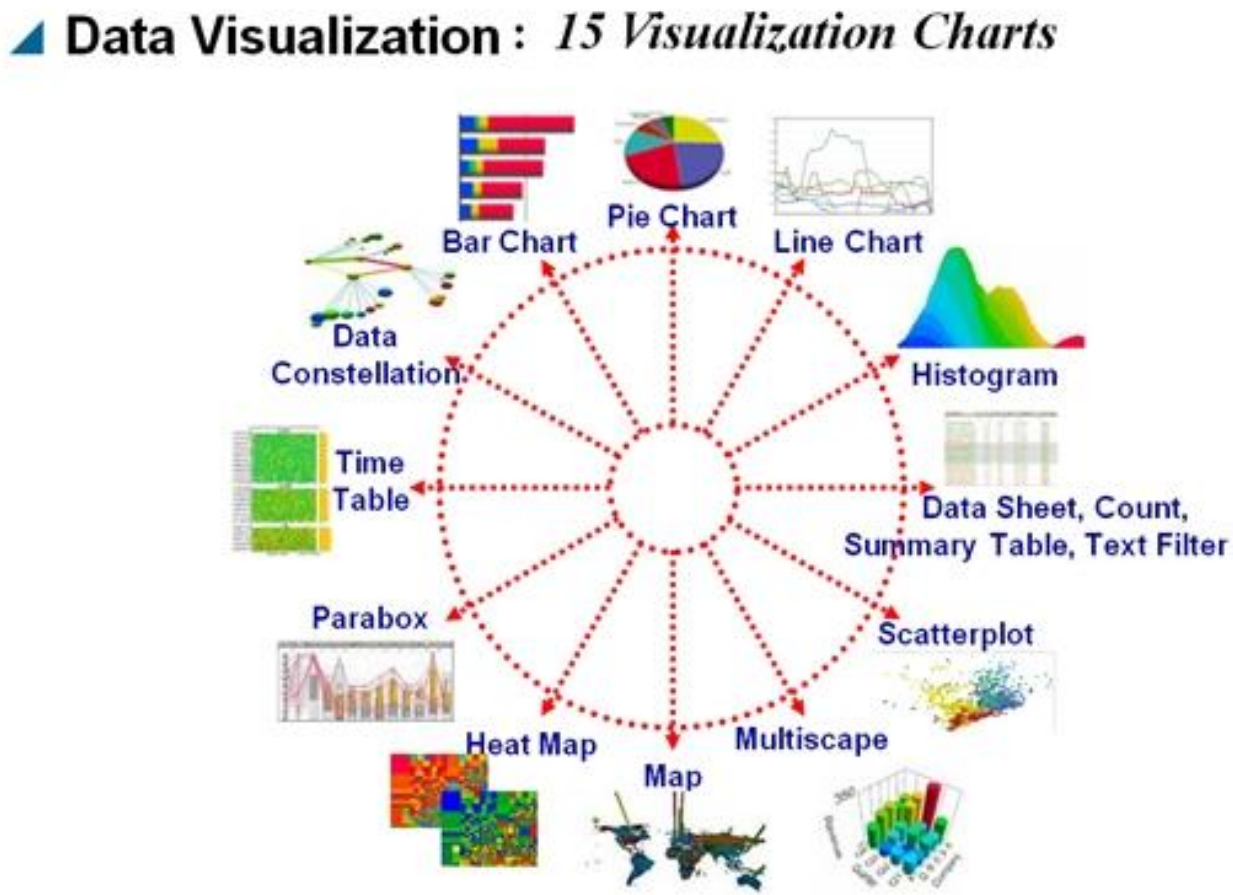
- As a BI software segment, visual data discovery tools are one of the most rapidly growing modules.

- Whenever there is a new category of software, there is confusion about what characterizes the software and what capabilities set it part.

- We could define visual data discovery as a **tool that speeds the time to insight through the use of visualizations, best practices in visual perception and easy exploration**.

- Such tools **support business agility and self-service BI through a variety of innovations that may include in-memory processing and mashing of multiple data sources**.

Visual Data Discovery



Source: https://visibledata.wordpress.com/2013/01/21/advizor-2/

Visual Data Discovery

- Visual data discovery moves the focus of BI from the "what" to exploring "why", "where" and "when".
- Users may not know precisely what information they are looing for and instead will navigate and drill within a data set to uncover particular details and patterns.
- For example:
  - In a business query tool, the question is **"What are my sales this month?"**
  - In a visual data discovery tool, the question may be **"What are the characteristics of those customers with higher sales?"** Perhaps it's age, income, gender or location.
- The results of the query may be rendered on a geographic map or scatter plot showing the relationship between sales, income and age.

Visual Data Discovery

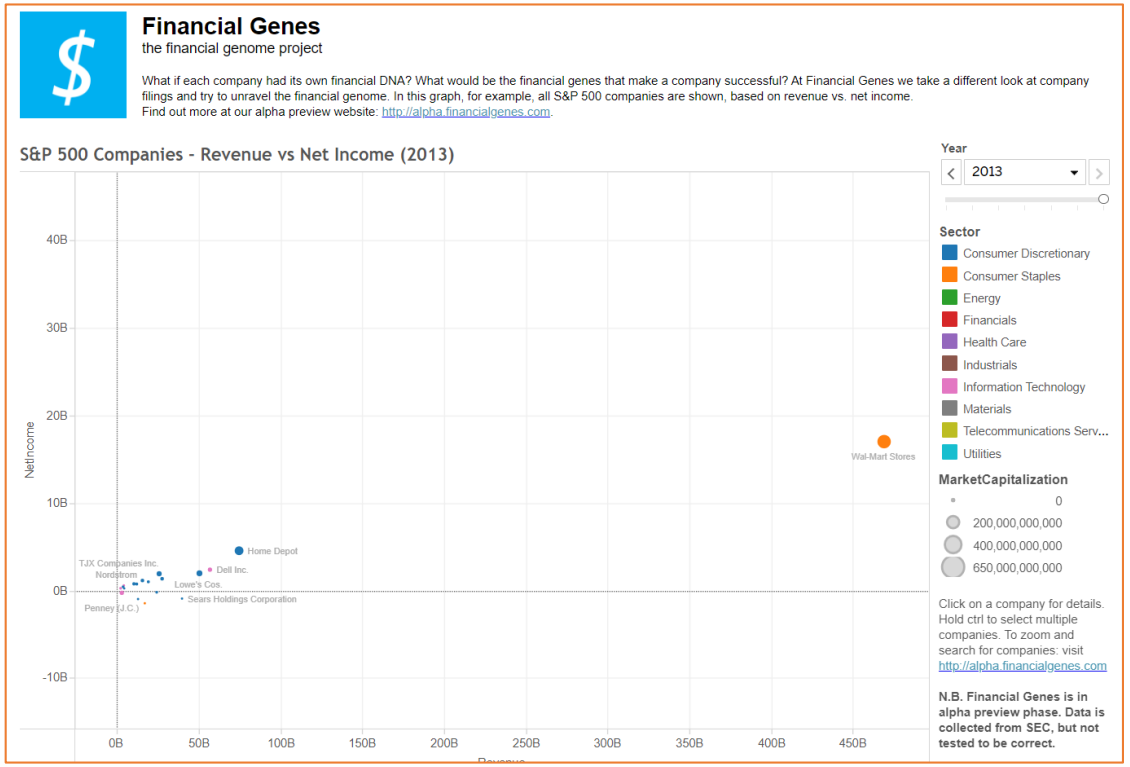
Comparison of Business Query and Visual Data Discovery Tools

	Business Query	Visual Data Discovery
Maturity	Mature, released in the early 1990s	Emerging
Business view	Required	Optional
IT involvement for initial setup	Yes to build business view metadata layer	Minimal, perhaps to create a query extract
Where authoring is done	Web	Mainly desktop, but some support Web
Initial data display	Tabular	Chart or tabular
Personal and multiple data sources	Rare but expanding	Common
Performance	Variable, depends on how well tuned the underlying database is	Fast, using in-memory processing, caching, or analytic database
Types of visualizations	Common bar, line, pie	Common types as well as tree maps, histogram, network, trellis, waterfall
Degree of interactivity for consumer	Initially and still often limited to refresh, but some vendors allow filter, sort, and drill on cached data set	Extensive and automatic filter, sort, lasso, drill, sliders, zoom

Visual Data Discovery

- Popular tools in this category include **Tableau**, TIBCO Spotfire, SAS Visual Analytics Explorer, MicroStrategy Visual Insight and SAP Lumira.

- Example:
  Visual data discovery for financial genome project using Tableau.



Source: https://public.tableau.com/app/profile/jasper.soetendal/viz/FinancialGenes-SP500KeyFinancials/Dashboard1

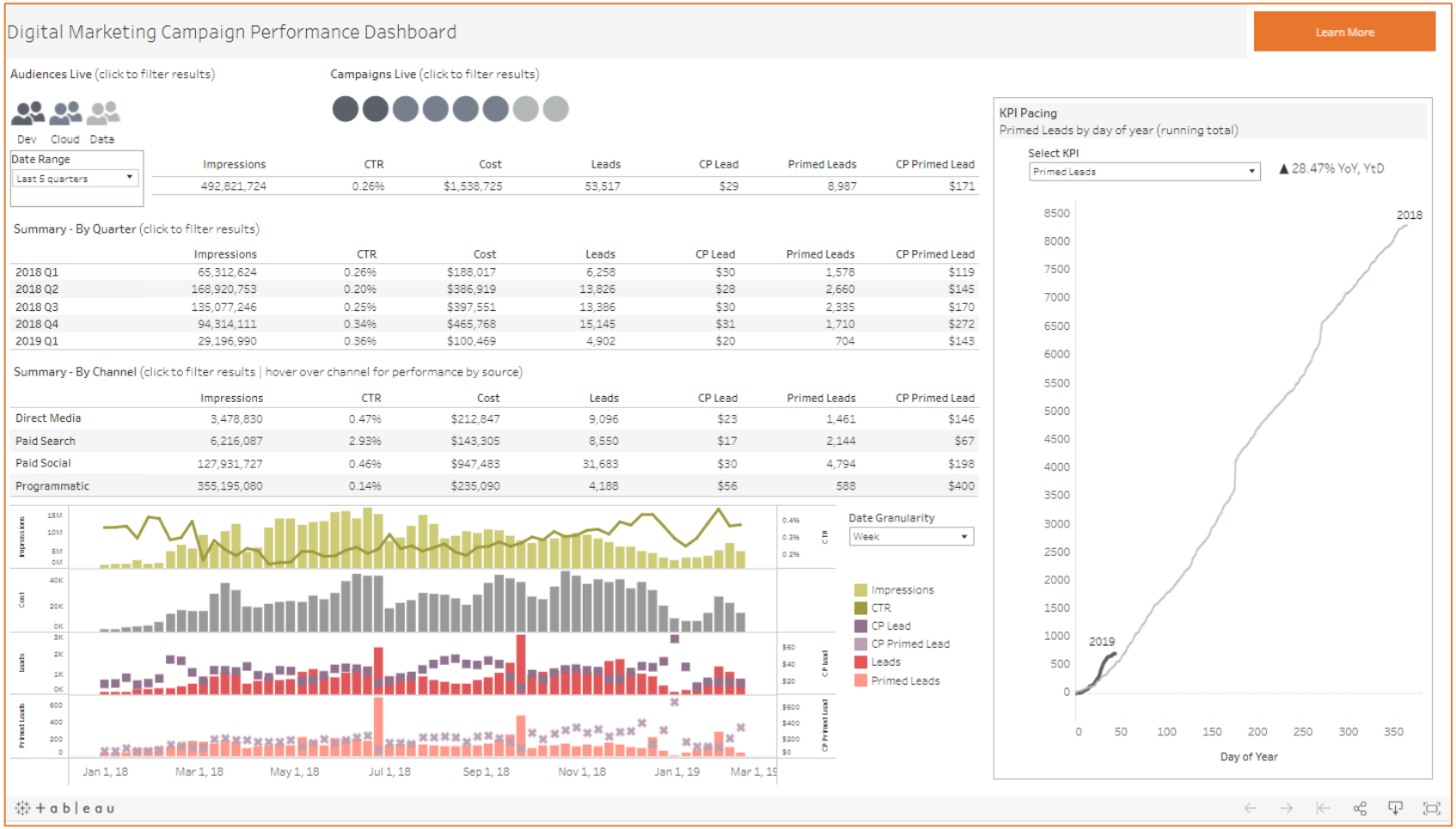
Dashboard

- Dashboard is a visual display of the most important information needed to achieve one or more objectives, consolidated and arranged on a single screen so the information can be monitored at a glance.

- A performance dashboard is a multi-layered application built on a business intelligence and data integration infrastructure that enables organizations to measure, monitor and manage business performance more effectively.

Dashboard

- **Example**: Performance dashboard created using Tableau.



Dashboard

- BI dashboards are similar to car dashboards – they provide multiple indicators or reports in a highly visual way.

- A dashboard may be composed of:
  - ✓ A map that color-codes where sales are performing well or poorly.
  - ✓ A trend line that tracks stock outs.
  - ✓ A cross tab of top-selling products.
  - ✓ A key performance indicator with an arrow to show if sales are according to plan.

- Ideally, users want to assemble their own dashboards with the information relevant to their job, and many visual data discovery tools allow for user-assembled dashboards.

Dashboard

- Other solutions are geared more for IT to design the dashboards. This is often preferred for operational dashboards that may access data in a source system (rather than a data warehouse), are refreshed in real time and support mission-critical operational tasks.

- A key characteristic of dashboards is that they present information from multiple data sources.

- Exactly how they do this and what constraints there are in the accessibility and number of data sources vary widely from product to product.

Scorecards

- The term “dashboards” and “scorecards” are often used interchangeably, although they are indeed different things.

- A major difference between them is that:
  - a **scorecard** tracks as set of key performance indicators (KPIs) compared to a forecast or target,
  - a **dashboard** will present multiple numbers in different ways.

- Some dashboards may additionally display metrics and target with visual traffic lighting to show the performance of that metric.

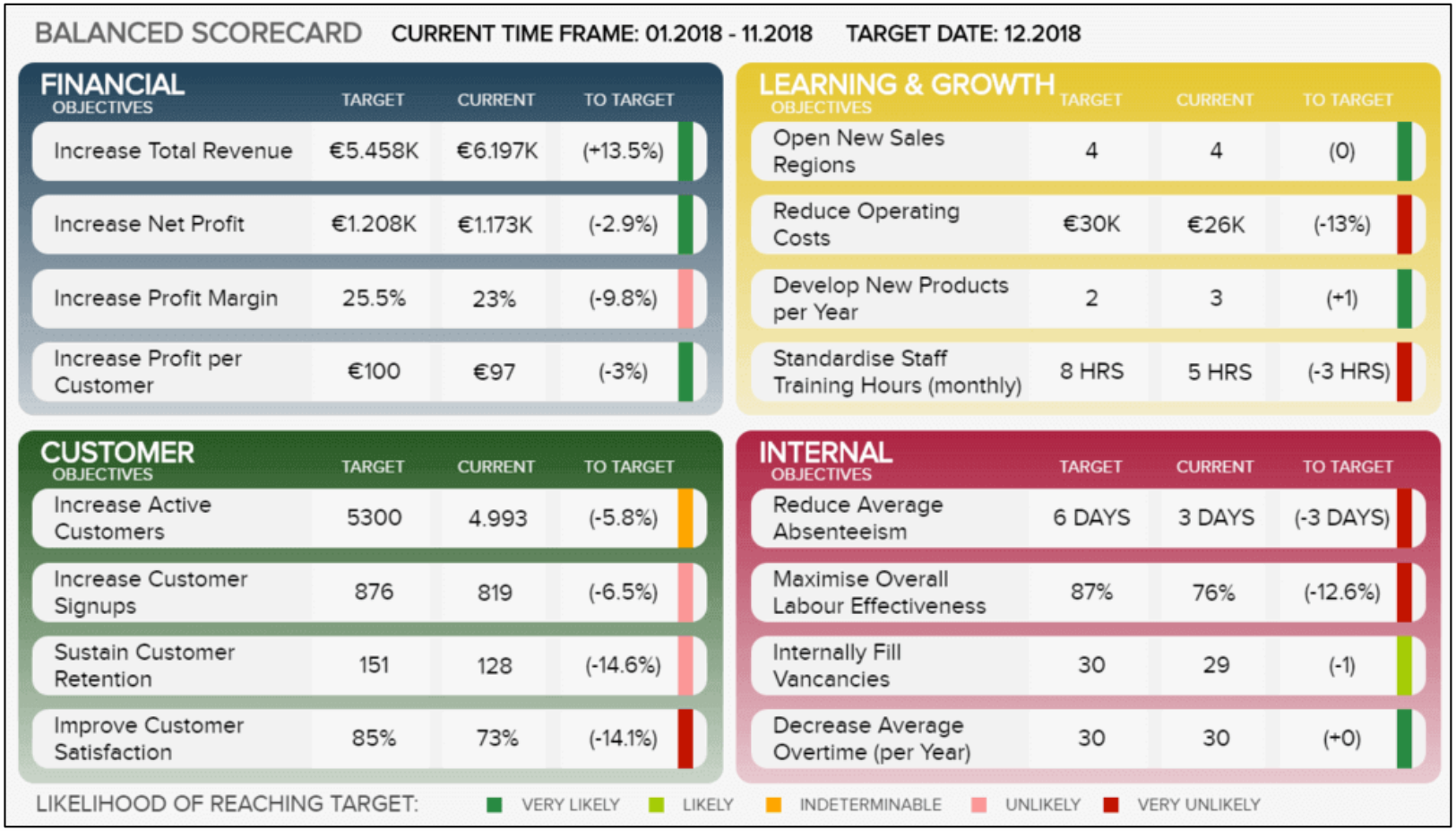
Scorecards

- Strategic scorecard contain metrics from the four key areas that drive the success of a business:
  - People
  - Customers
  - Financial
  - Operations

and will include strategy maps to show how the metrics relate to one another.

Scorecards

- **Example**: Balanced Scorecard



BALANCED SCORECARD CURRENT TIME FRAME: 01.2018 - 11.2018 TARGET DATE: 12.2018

FINANCIAL OBJECTIVES	TARGET	CURRENT	TO TARGET
Increase Total Revenue	€5.458K	€6.197K	(+13.5%)
Increase Net Profit	€1.208K	€1.173K	(-2.9%)
Increase Profit Margin	25.5%	23%	(-9.8%)
Increase Profit per Customer	€100	€97	(-3%)

LEARNING & GROWTH OBJECTIVES	TARGET	CURRENT	TO TARGET
Open New Sales Regions	4	4	(0)
Reduce Operating Costs	€30K	€26K	(-13%)
Develop New Products per Year	2	3	(+1)
Standardise Staff Training Hours (monthly)	8 HRS	5 HRS	(-3 HRS)

CUSTOMER OBJECTIVES	TARGET	CURRENT	TO TARGET
Increase Active Customers	5300	4.993	(-5.8%)
Increase Customer Signups	876	819	(-6.5%)
Sustain Customer Retention	151	128	(-14.6%)
Improve Customer Satisfaction	85%	73%	(-14.1%)

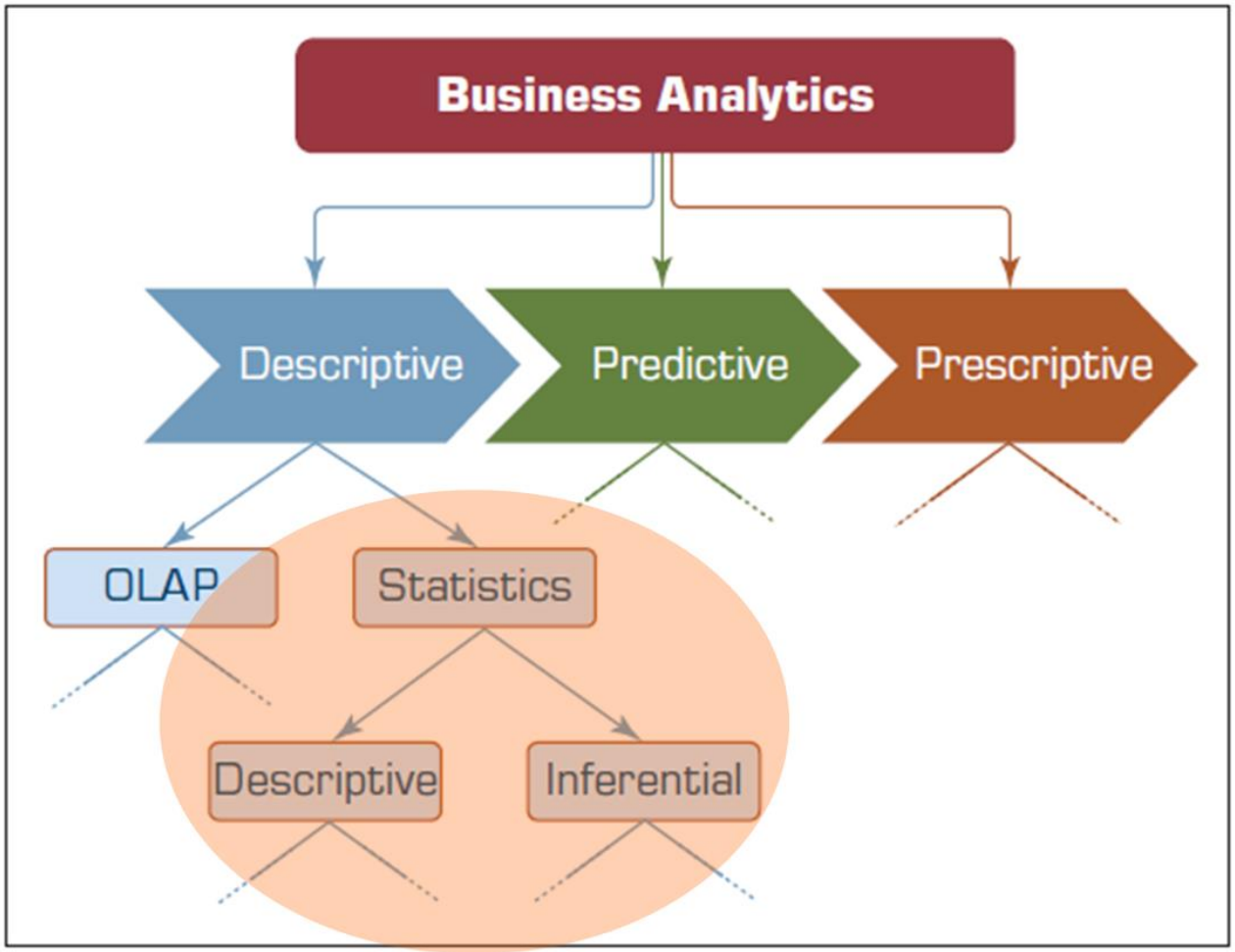
INTERNAL OBJECTIVES	TARGET	CURRENT	TO TARGET
Reduce Average Absenteeism	6 DAYS	3 DAYS	(-3 DAYS)
Maximise Overall Labour Effectiveness	87%	76%	(-12.6%)
Internally Fill Vancancies	30	29	(-1)
Decrease Average Overtime (per Year)	30	30	(+0)

LIKELIHOOD OF REACHING TARGET: VERY LIKELY LIKELY INDETERMINABLE UNLIKELY VERY UNLIKELY

Business Analytics



Statistical Modeling for Business Analytics



Statistical Modeling for Business Analytics

- **Statistics**
  - A collection of mathematical techniques to characterize and interpret data.
- **Descriptive Statistics**
  - Describing the data (as it is).
- **Inferential statistics**
  - Drawing inferences about the population based on sample data.

Descriptive Statistics

Descriptive Statistics (DS) for Descriptive Analytics:

- Using simple formulas, numerical aggregations, data tables and charts.
- Using numbers and symbols to summarises data to represent it.
- Often meaningful patterns emerge.
- Help to characterise and validate data.
- Does not allow making conclusions/inferences beyond sample of data.

Descriptive Statistics:
Measures of Centrality Tendency

- **Arithmetic Mean**

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} \quad \Rightarrow \quad \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- **Median**
  - The number in the middle.
- **Mode**
  - The most frequent observation.

Descriptive Statistics:
Measures of Dispersion

- **Dispersion**
  - Degree of variation in a given variable
- **Range**
  - Max - Min
- **Variance**

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

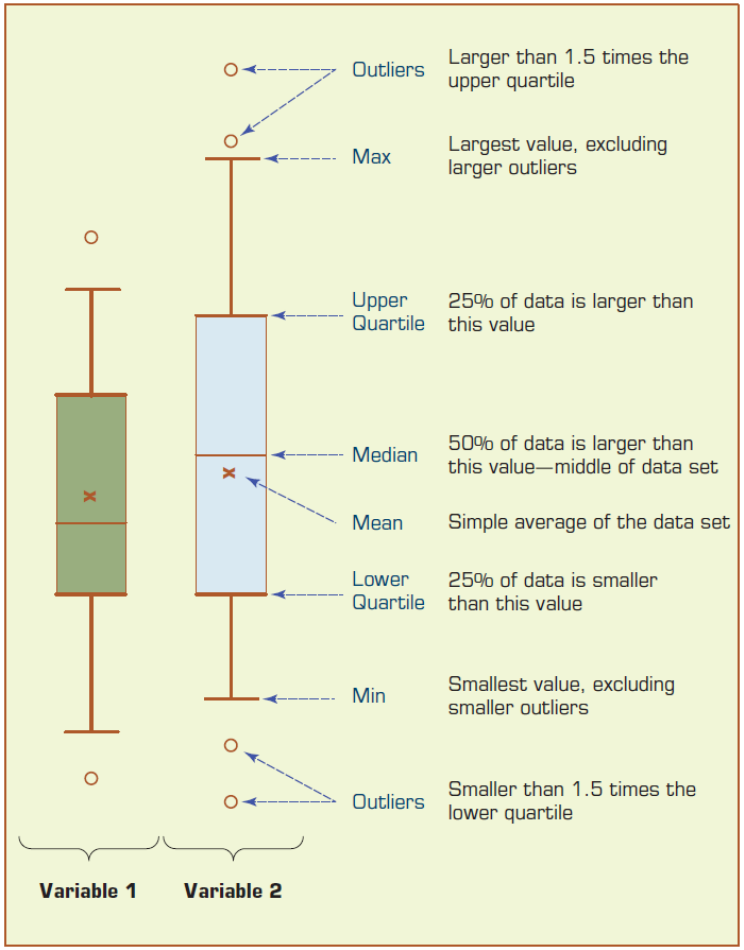
- **Standard Deviation**

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

- **Mean Absolute Deviation (MAD)**
  - Average absolute deviation from the mean

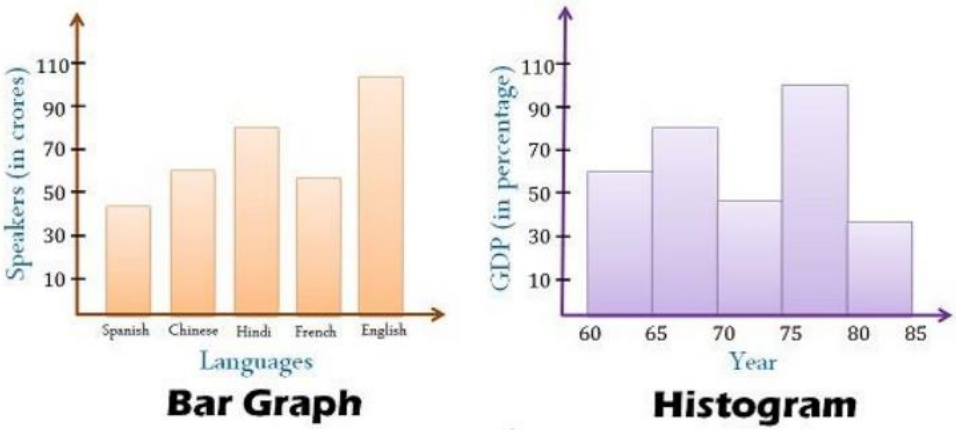
Descriptive Statistics:
Measures of Dispersion

- **Quartiles**
- **Box-and-Whiskers Plot:**
  - ✓a.k.a. box-plot
  - ✓Versatile / informative



Descriptive Statistics:

Histogram and Bar Graphs



- A histogram shows distribution of a variable typically on axis-x. If the variable is of quantitative, it is binned.
- Bar graph shows comparison and contains multiple variables that should/could be reordered.

Descriptive Statistics:
Measures of Dispersion - Shape Properties

- **Histogram** – frequency chart
- **Skewness**
  - Measure of asymmetry

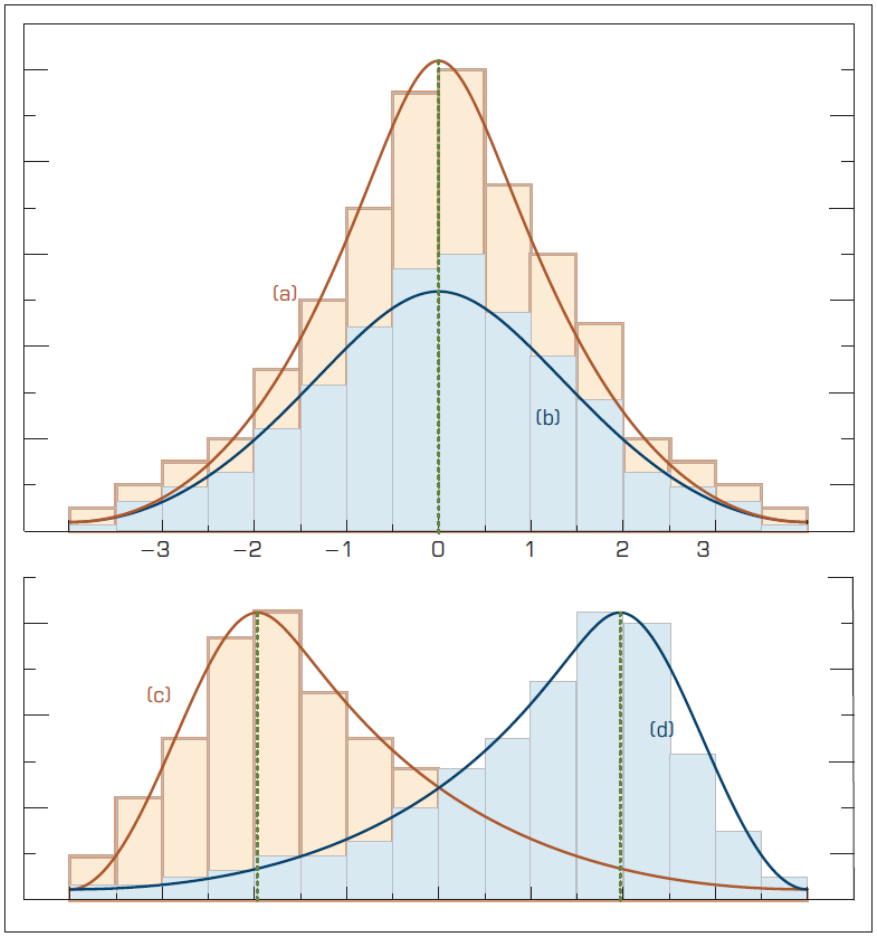
$$Skewness = S = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3}{(n-1)s^3}$$

- **Kurtosis**
  - Peak/tall/skinny nature of the distribution

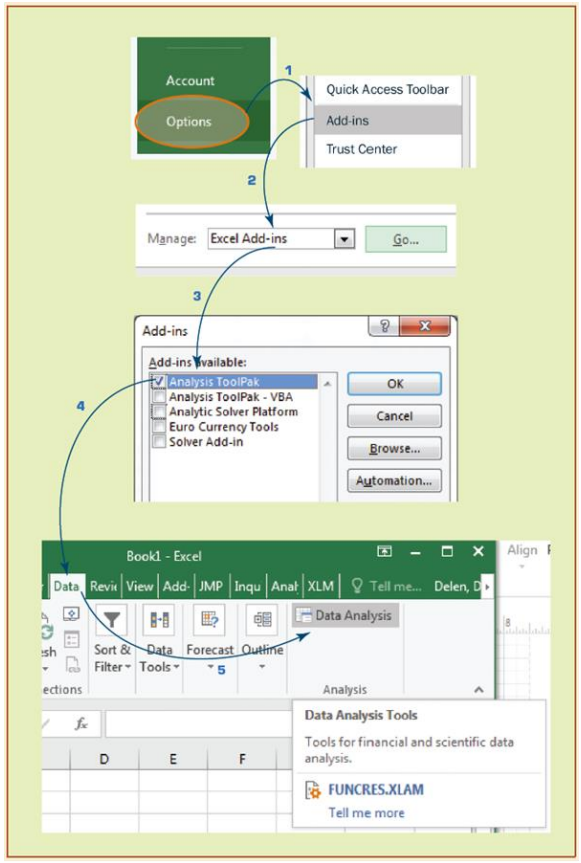
$$Kurtosis = K = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^4}{ns^4} - 3$$

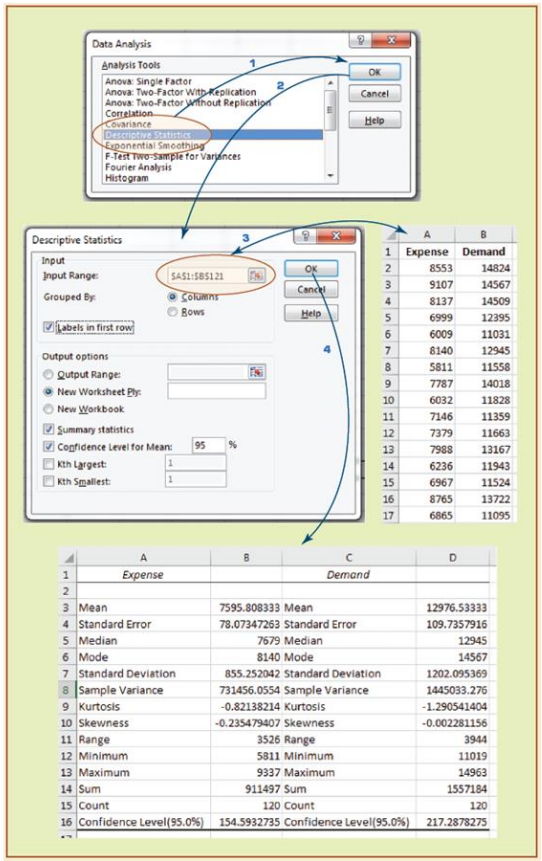
Descriptive Statistics:

Relationship between Dispersion & Shape Properties



Technology Insight:
Descriptive Statistics in MS Excel

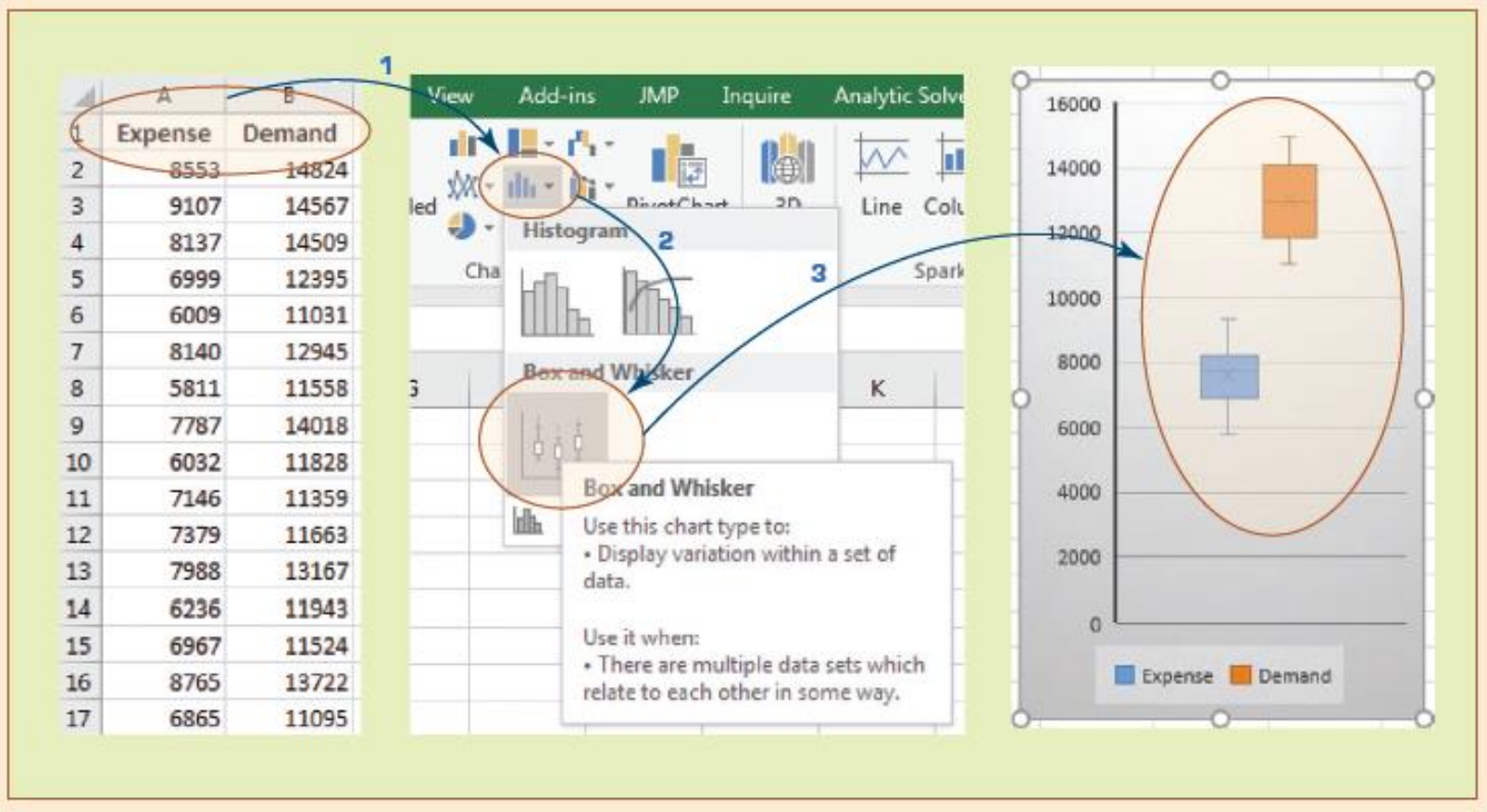




	A	B
1	Expense	Demand
2	8553	14824
3	9107	14567
4	8137	14509
5	6999	12395
6	6009	11031
7	8140	12945
8	5811	11558
9	7787	14018
10	6032	11828
11	7146	11359
12	7379	11663
13	7988	13167
14	6236	11943
15	6967	11524
16	8765	13722
17	6865	11095

	A	B	C	D
1	*Expense*		*Demand*	
2				
3	Mean	7595.808333	Mean	12976.53333
4	Standard Error	78.07347263	Standard Error	109.7357916
5	Median	7679	Median	12945
6	Mode	8140	Mode	14567
7	Standard Deviation	855.252042	Standard Deviation	1202.095369
8	Sample Variance	731456.0554	Sample Variance	1445033.276
9	Kurtosis	-0.82138214	Kurtosis	-1.290541404
10	Skewness	-0.235479407	Skewness	-0.002281156
11	Range	3526	Range	3944
12	Minimum	5811	Minimum	11019
13	Maximum	9337	Maximum	14963
14	Sum	911497	Sum	1557184
15	Count	120	Count	120
16	Confidence Level(95.0%)	154.5932735	Confidence Level(95.0%)	217.2878275

Technology Insight:
Descriptive Statistics in MS Excel



Technology Insight:

Descriptive Statistics in others (software/tools)

- Tableau
- Power BI
- Alteryx
- SPSS
- SAS
- R
- Python
- ….